EXPERIMENT NO. 02—Line coding Techniques



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TITLE: Design of Line Coding Techniques.

PROBLEM STATEMENT: Using the PN sequence generated in the previous experiment as data:

1. Design and implement a circuit for the working of Manchester Line coding scheme.
2. Design and implement a circuit showing the working of Alternate Mark Inversion (AMI).
3. Implement an MLT-3 line coding technique in MATLAB for a stream of bits of digital data.

KEY COMPONENTS: IC7486(XOR gate), IC741(Op-amp), IC74164(8-bit shift register), IC7474(D Flip Flop), digital oscilloscope, function generator, breadboard, connecting wires, power supply.

BRIEF THEORY & BACKGROUND:
The objective of the experiment is to implement various line-coding techniques by designing circuits. In digital communication systems, when the message is source encoded, the digital output of that source encoder is coded into electrical pulses which are then transmitted over the channel. This process is called line encoding [1].

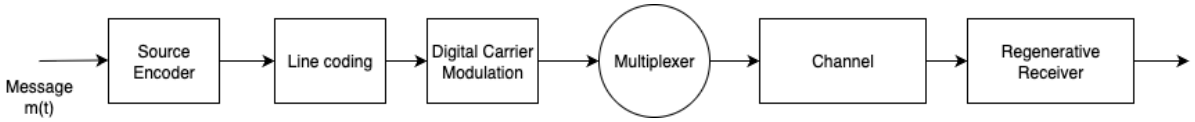


Figure 1: Line coding in Digital Communication Systems [1]

Line coding helps in reducing error rate during transmission, enablimg reliable data transmission, increase in bandwidth efficiency and facilitating error detection and correction [2].

Different Types of line Encoding are:

1. **On–Off (Unipolar RZ)** - Here 1 is represented by pulse $p(t)$ and 0 is represented as no pulse. It is the simplest form of line coding. For 1, for its half time interval the pulse $p(t)$ returns to zero.
2. **Polar(RZ)** - Here 1 is represented by pulse $p(t)$ and 0 is represented by $-p(t)$. It is the most power efficient code.
3. **Bipolar(RZ)(Alternate Mark Inversion/Pseudoternary)** - here 0 is represented by no pulse and 1's are represented by alternating $p(t)$ and $-p(t)$. For expample if the first 1 has $p(t)$ then the next 1 will have $-p(t)$. It provides error detection capability when the alternation rule is violated.
4. **On-Off (RZ)** - Here for the entire suration of 1, pulse $p(t)$ does not return to zero, but for 0, it returns to zero.

5. **Polar (NRZ)** - Here for 1 pulse has $p(t)$ and for 0, it has $-p(t)$.
   [1]

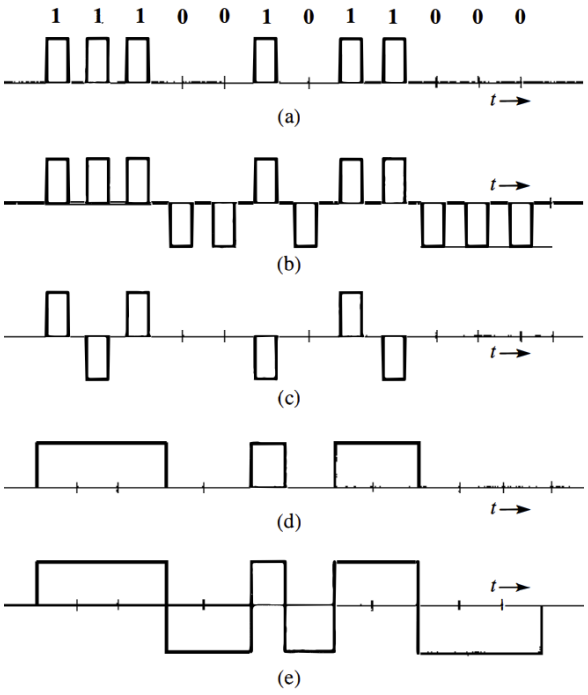


Figure 2: Different line coding techniques. (a) On–Off (Unipolar RZ), (b) Polar(RZ), (c) Bipolar(RZ)(Alternate Mark Inversion/Pseudoternary), (d) On-Off (RZ), (e) Polar (NRZ)
[1]

Advantages/Disadvantages of each Line coding scheme and their core usage:

1. On–Off (Unipolar RZ) - Advantages include being simple and lesser bandwidth being used. Major disadvantages include no error correction, possibility of synchronization loss[3]. Its usage include Morse Code and ISM bands to transfer data between computers[4].

2. Polar (RZ) - Advantages include being simple and eliminates low-frequency components. However disadvatages include not providing error correction and requires twice the bandwidth of Polar NRZ[3].

3. Bipolar (RZ) - Advantages include being simple, having no low-frequency components, using less bandwidth than unipolar and polar NRZ and allowing single error detection. Major disadvantage is that if long strings of data are there, it can lead to loss of synchronization. It is mainly used in telephone networks and older digital communication systems[3].

4. On-Off (NRZ) - It is simple and requires less bandwidth, but it does not provide error correction, and the presence of low-frequency components may cause signal droop[3].

5. Polar (NRZ) - It is simple and does not contain low-frequency components, but it does not offer error correction[3].

OUTCOMES:

Manchester Coding :

In Manchester coding, the bits 0 and 1 are represented by 1 to 0 (or high to low) transition and 0 to 1 (or low to high transition) or vice versa [5].

The following is the data in DSO:



Figure 3: Manchester coding. (a) Plot C1 (PN Sequence), (b) Plot C2 Manchester Coded Sequence. Note the transition from low to high in 1 and high to low in 0

Circuit diagram for Manchester Coding was drawn in LTSpice. Hardware implementation was done in Lab. Following is the circuit diagram:

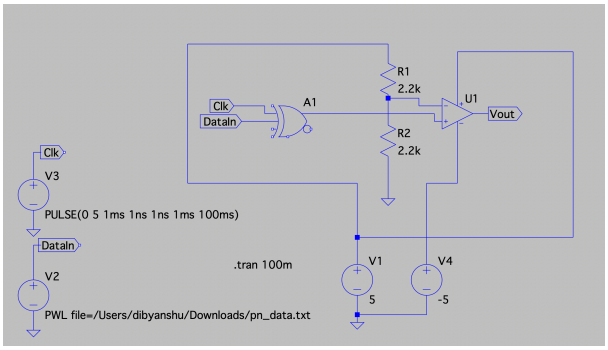


Figure 4: Manchester coding LTSpice implementation. Note :No simulation in LTSpice was done. This is a diagramatic representation.

AMI Coding:

AMI (or Alternate Mark Inversion) coding represents a coding technoque where every alternate 1's is represented by pulses of opposite polarity. Suppose if we have 1 and 1 back to bak, the first 1 will be represented by positive pulse and the next one will be represented by a pulse of opposite polarity. 0 will be represented by 0[6].

The following is the DSO data for AMI.

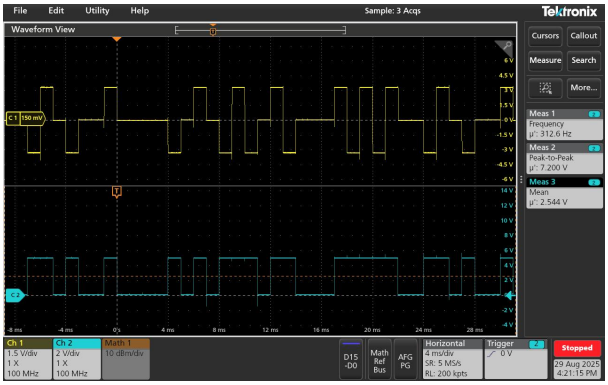


Figure 5: AMI DSO data. C1 is the AMI data whereas C2 is the PN Sequence generated.

The following is the Circuit implementation in LTSpice.

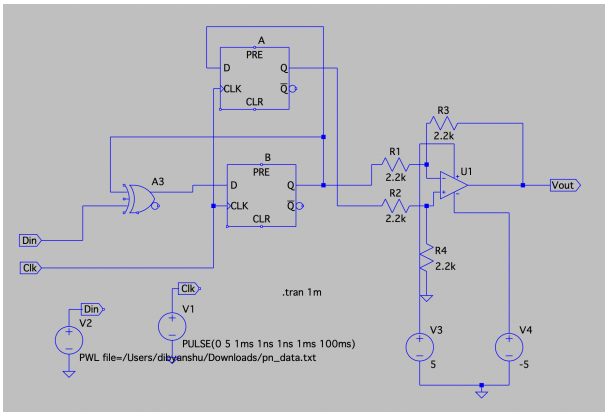


Figure 6: AMI Circuit Diagram. No simulation was done in LTSpice. This is only to represent the overall circuitry AMI done in lab.

The following is the MATLAB plot for AMI code generated.

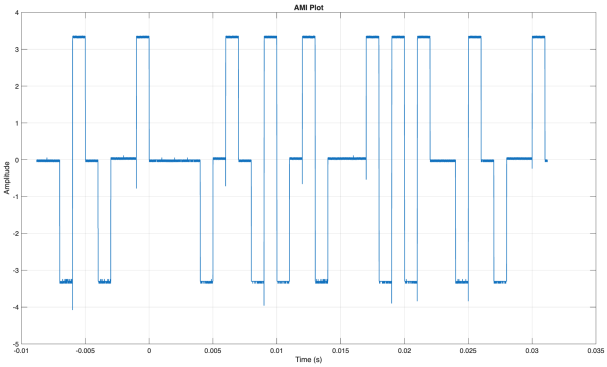


Figure 7: MATLAB Plot for AMI Coding.

The following is the state diagram of the AMI Coding technique.

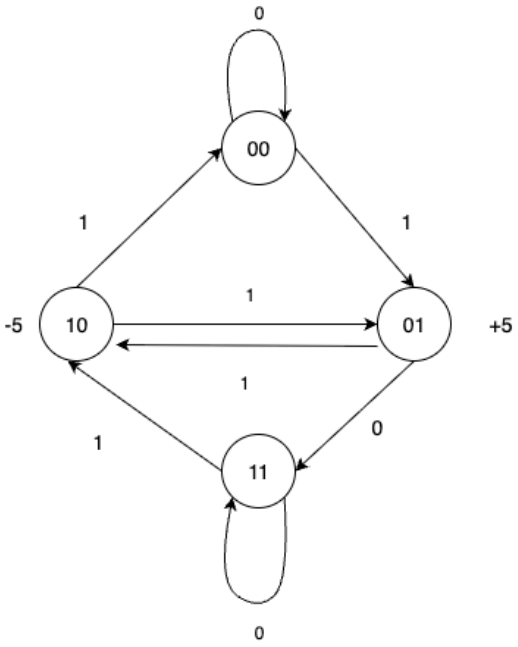


Figure 8: Note that 01 and 10 are chosen to represent bipolar volatges..

Based on the above state diagram the following table is obtained to represent the next states based on input and present state.

AMI State Transition Table

A_n	B_n	D	A_{n+1}	B_{n+1}
0	0	0	0	0
0	0	1	0	1
0	1	0	1	1
0	1	1	1	0
1	1	0	1	1
1	1	1	1	0
1	0	0	0	0
1	0	1	0	1

MLT3 Coding:
In MLT3 encoding there are 3 voltage levels that are used - +v, 0, -V. it follows the following pattern: - If the next bit is 0, there will be no transition; if the next bit is 1 and current level is not 0, then the next level will be 0; if the next bit is 1 and the current bit is 0, then the next bit will be the opposite of the previous non zero level [7].

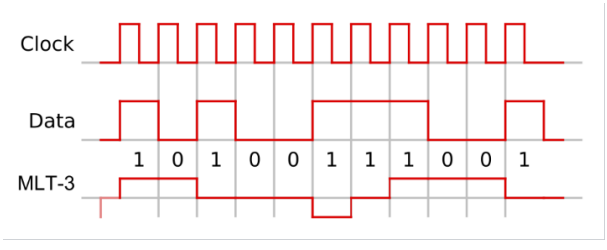


Figure 9: MLT3 Encoding Representation.[4]

The following figure displays the state diagram for MLT3 Encoding. The above logic is being shown here [8]:

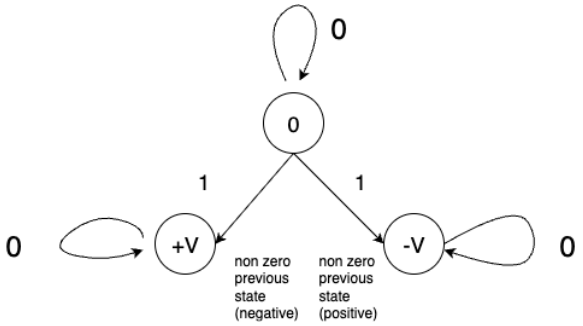


Figure 10: State Diagram of MLT3 Encoding.

The following is the MATLAB graph for MLT3 encoding.

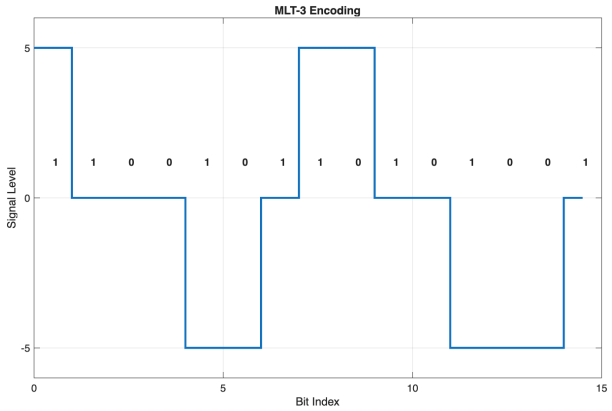


Figure 11: MLT3 Encoding in Matlab.

MATLAB CODE for MLT3 encoding:

```
clc;
clear;
close all;


bits = [1 1 0 0 1 0 1 1 0 1 0 1 0 0 1]; % Input 11 bit sequence

% Initialize
level = 0;  % level iniotialisation
mlt3 = [];  % initialising mtl3 as a vector
next_pol = 5;     % next_polar to check the polarity of the sequence and
    convert it as such (+5 taken)

for i = 1:length(bits)
    if bits(i) == 0 % checks if the bit is 0
        level = level; % no change in level
    elseif bits(i) == 1
        % For '1', move in the cycle
```

```
        if level == 0 % checks if the current level is zero
            level = next_pol; % if yes then change that level to next
                polarity
            next_pol = -next_pol; % flip polarity for next time
        else
            level = 0;
        end
    end
    % Repeat each level for one bit duration (two samples per bit for
        stairs plot)
    mlt3 = [mlt3 level level];
end

% Time axis (two samples per bit)
t = 0:0.5:length(bits)-0.5;

% Plot waveform
stairs(t, mlt3, 'LineWidth', 2); % for smooth rectangular graph
ylim([-6 6]); grid on; % putting limit on graph
xlabel('Bit Index');
ylabel('Signal Level');
title('MLT-3 Encoding');
set(gca,'YTick',[-5 0 5]); % having values of -5 0 and 5

% Show input bits above waveform
for i = 1:length(bits)
    text(i-0.5, 1.2, num2str(bits(i)), 'FontWeight','bold');
end
```

DISCUSSION:

1. I was able to understand NRZ, RZ, bipolar and polar encoding properly.
2. In the experiment, first we had to generate a PN Sequence. This was (and is) a challenging part for me, but with the help of my group member (Akash), I was able to do it.
3. The next was to understand how to take the data input and get the Manchester code. This was done by Using Clock and PN Sequence generated output and suing XOR operation on it and I was able to understand how Manchester code is generated.
4. The next challenge was how to make sure that the data in the output was between +5 V and -5 V. This I was not able to do. akash helped me in understanding the use of Op-Amp, designing a voltage divider in hardware and use of IC741 op amp.
5. AMI was tough for me to understand as I was finding it difficut to understand the state diagram. Akash helped me understand it and I was able to get the state diagram. We were also able to design the circuit in hardware by using XOR gate, D flip, flop and Op-Amp. Designing Op-amp was tricky but akash patiently corrected me multiple times.
6. I was able to use LTSpice for the first time to draw the circuit diagram properly for Manchester and AMI coding. However I was not able to simulate the results. More study is needed there.
7. Using LLM, I was able to understand the logic of MTL3 Encoding and generate the MATLAB code. However it took me considerable time to understand the code and I tried to write it myself. Still considerable studying needs to be done here.

8. I was also able to draw the state diagrams using draw.io, thanks to my classmate Sai Dinesh.

9. There is still a need to undersatnd the 2 level AMI code and k level AMI code, which I will do in future.

References

[1] B. P. Lathi and Z. Ding, "Principles of digital data transmission," in *Modern Digital and Analog Communication Systems*, 4th ed., New York, NY, USA: Oxford University Press, 2010, ch. 7, p. 327.

[2] "Line coding in digital communication." [Online; accessed 5-September-2025], Studytonight. [Online]. Available: https://www.studytonight.com/post/line-coding.

[3] "Digital communication – line codes." [Online; accessed 5-September-2025], Tutorialspoint. [Online]. Available: https://www.tutorialspoint.com/digital_communication/digital_communication_line_codes.htm.

[4] Wikipedia contributors, *On–off keying — Wikipedia, the free encyclopedia*, [Online; accessed 5-September-2025], 2024. [Online]. Available: https://en.wikipedia.org/w/index.php?title=On%E2%80%93off_keying&oldid=1233030851.

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[6] G. Fairhurst, *Alternate mark inversion (ami)*, https://erg.abdn.ac.uk/users/gorry/course/phy-pages/ami.html, [Accessed: 07-Sep-2025], Nov. 2020.

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